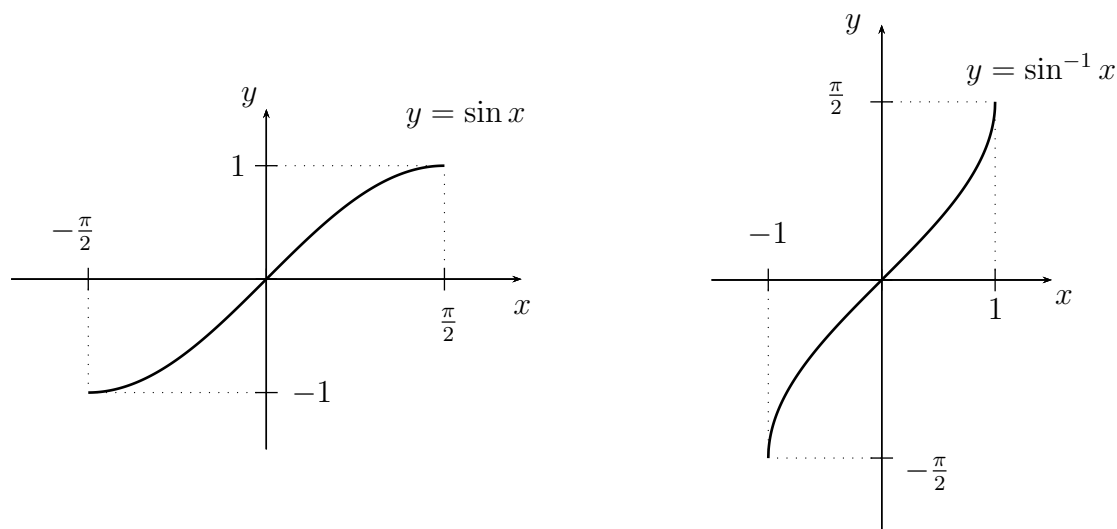


Notes for 'Inverse circular functions'

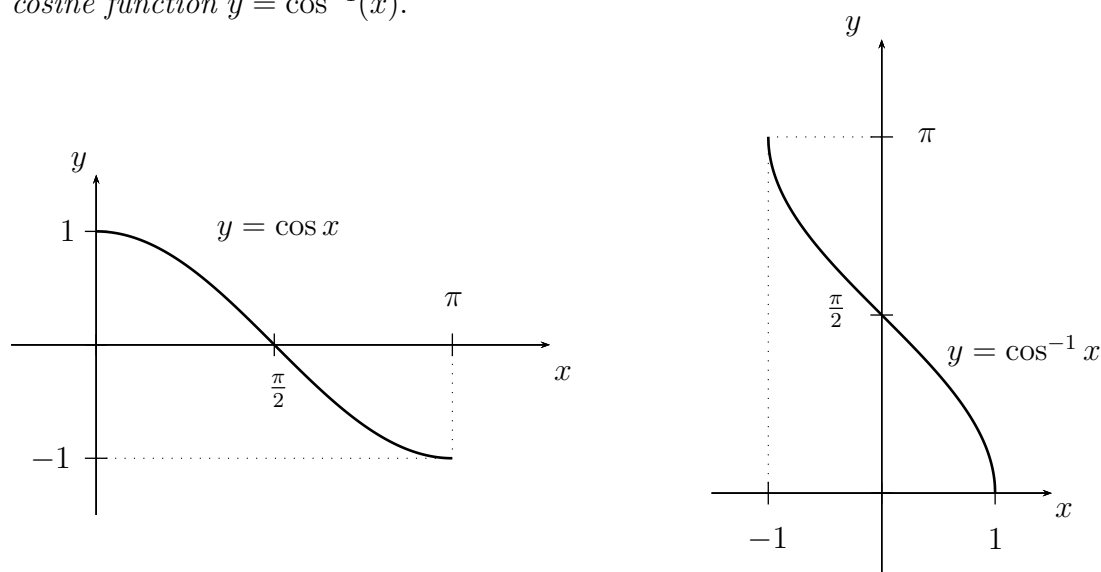
Important Ideas and Useful Facts:

- (i) **The inverse sine function:** It is conventional to restrict the domain of $y = \sin x$ to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ to produce a fragment of the sine curve that satisfies the horizontal line test. By reflecting this fragment in the line $y = x$, we produce the graph of the *inverse sine function* $y = \sin^{-1}(x)$.



The domain of $\sin^{-1}$ is the interval $[-1, 1]$ and the range is the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

- (ii) **The inverse cosine function:** It is conventional to restrict the domain of $y = \cos x$ to the interval $[0, \pi]$ to produce a fragment of the cosine curve that satisfies the horizontal line test. By reflecting this fragment in the line $y = x$, we produce the graph of the *inverse cosine function* $y = \cos^{-1}(x)$.

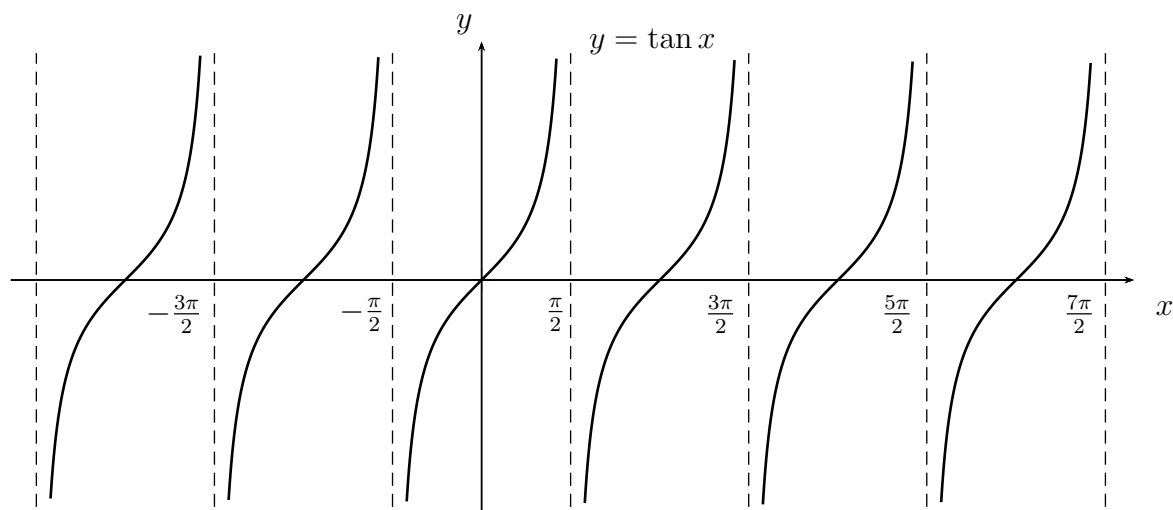


The domain of $\cos^{-1}$ is the interval $[-1, 1]$ and the range is the interval $[0, \pi]$.

(iii) The tangent function: Recall that the *tangent function*

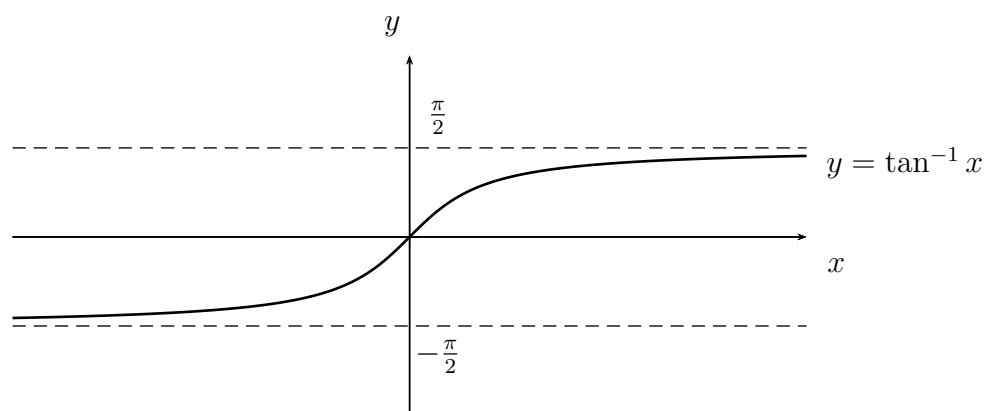
$$y = \tan x = \frac{\sin x}{\cos x}$$

is only defined when $\cos x$ is nonzero. In particular, $\cos(\frac{\pi}{2}) = \cos(\frac{-\pi}{2}) = 0$, so that $\tan(\frac{\pi}{2})$ and $\tan(\frac{-\pi}{2})$ are both undefined. In fact, the lines $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$ form vertical asymptotes.



The periodic behaviour of the sine and cosine functions induce periodic behaviour in the tan function, and vertical asymptotes appear along the x -axis at every odd multiple of $\frac{\pi}{2}$.

(iv) The inverse tangent function: It is conventional to restrict the domain of $y = \tan x$ to the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ to produce the branch of the tangent curve above sitting between the asymptotes $x = \pm\frac{\pi}{2}$, which certainly satisfies the horizontal line test. By reflecting this branch in the line $y = x$, we produce the graph of the *inverse tangent function* $y = \tan^{-1}(x)$, which has a *sigmoid shape* and horizontal asymptotes $y = \pm\frac{\pi}{2}$. The domain of the inverse tangent function is all of the real line $\mathbb{R}$, and the range is the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.



Examples:

1. We can think of the sine, cosine and tangent functions as taking angles and producing numbers. Therefore we can think of the inverse sine, cosine and tangent functions as bringing numbers back to angles. For example,

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}, \quad \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \quad \sin\left(\frac{\pi}{2}\right) = 1, \quad \sin\left(-\frac{\pi}{2}\right) = -1,$$

so “undoing” each of these gives

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}, \quad \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}, \quad \sin^{-1}(1) = \frac{\pi}{2}, \quad \sin^{-1}(-1) = -\frac{\pi}{2}.$$

For example,

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}, \quad \cos\left(\frac{\pi}{2}\right) = 0, \quad \cos(\pi) = -1,$$

so “undoing” each of these gives

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}, \quad \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}, \quad \cos^{-1}(0) = \frac{\pi}{2}, \quad \cos^{-1}(-1) = \pi.$$

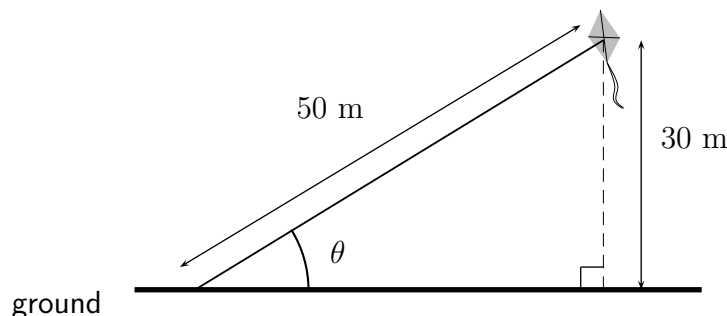
For example,

$$\tan\left(\frac{\pi}{4}\right) = 1, \quad \tan\left(-\frac{\pi}{4}\right) = -1, \quad \tan\left(\frac{\pi}{3}\right) = \sqrt{3}, \quad \tan\left(-\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}},$$

so “undoing” each of these gives

$$\tan^{-1}(1) = \frac{\pi}{4}, \quad \tan^{-1}(-1) = -\frac{\pi}{4}, \quad \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}, \quad \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}.$$

2. A kite is attached to the ground by a piece of string of length 50 metres. The kite is flying 30 metres directly above the ground, and we will assume that the string is approximately straight. The problem is to find the angle θ of elevation.



Solution: We have a right-angled triangle with angle θ , opposite side length 30 m and hypotenuse 50m. Hence

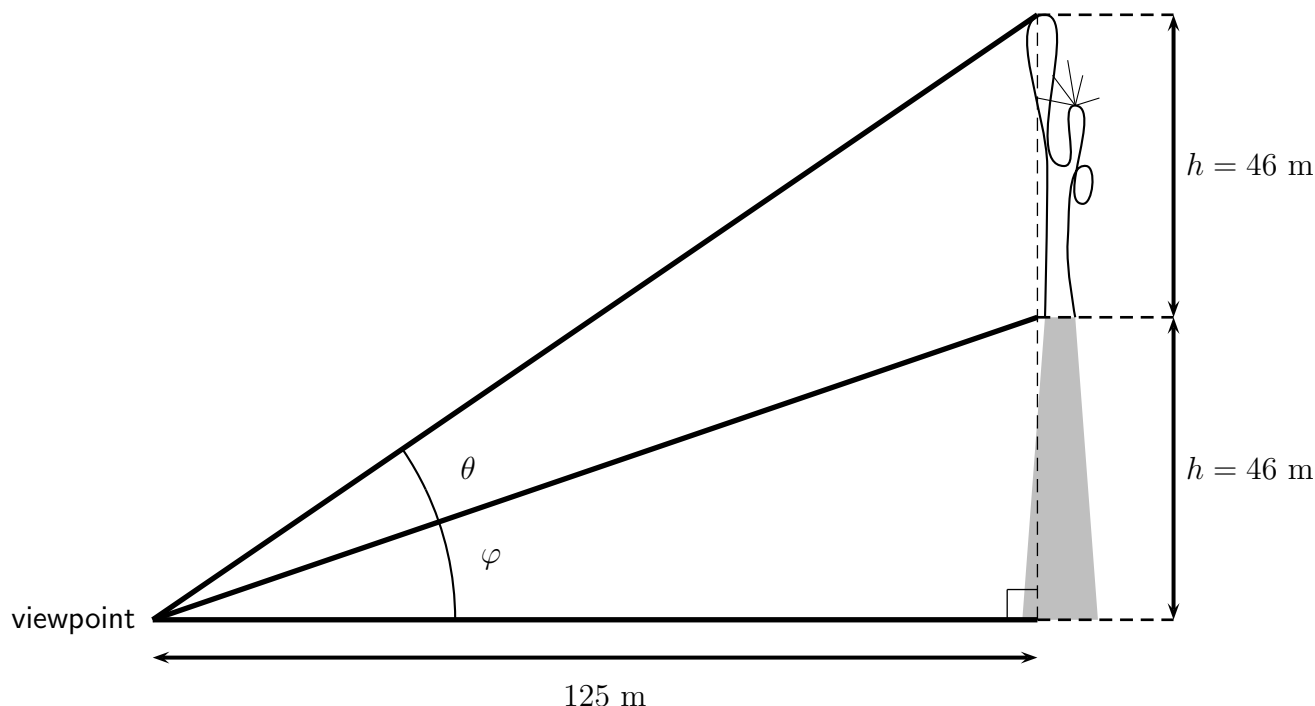
$$\sin \theta = \frac{30}{50} = 0.6,$$

so, “undoing” the sine function, we get

$$\theta = \sin^{-1}(0.6) \approx 0.6435 \approx 37^\circ.$$

We estimate the angle of elevation of the kite to be about 37° .

3. The Statue of Liberty and its pedestal are both 46 metres tall.



The problem is to find the angle θ subtended by the Statue when viewed from a horizontal distance 125 metres from the base of the pedestal.

Solution: Let φ be the angle subtended by the pedestal, so

$$\tan \varphi = \frac{46}{125} \quad \text{and} \quad \tan(\varphi + \theta) = \frac{46 + 46}{125} = \frac{92}{125}.$$

Hence, “undoing” the tan function gives

$$\varphi = \tan^{-1}\left(\frac{46}{125}\right) \quad \text{and} \quad \varphi + \theta = \tan^{-1}\left(\frac{92}{125}\right),$$

so that

$$\theta = \tan^{-1}\left(\frac{92}{125}\right) - \varphi = \tan^{-1}\left(\frac{92}{125}\right) - \tan^{-1}\left(\frac{46}{125}\right) \approx 0.28186 \approx 16^\circ.$$

Thus the angle subtended by the Statue of Liberty from 125 metres from the base is approximately 16° .